PORTI LAB 2

Lab 02: Network Security Groups and Application Security Groups (NSG + ASG)

Region: West US 2
Resource Group: `FortisAid-Lab02`
Goal: Use **ASGs** to group VMs logically and use a **subnet-level NSG** to enforce least-privilege inbound access:

- Web tier: Allow **HTTP/HTTPS (80/443)**, block **RDP (3389)**
- Mgmt tier: Allow **RDP (3389)**, no web exposure required

Exercise 1: Create Virtual Networking Infrastructure

Step 1 — Create Resource Group

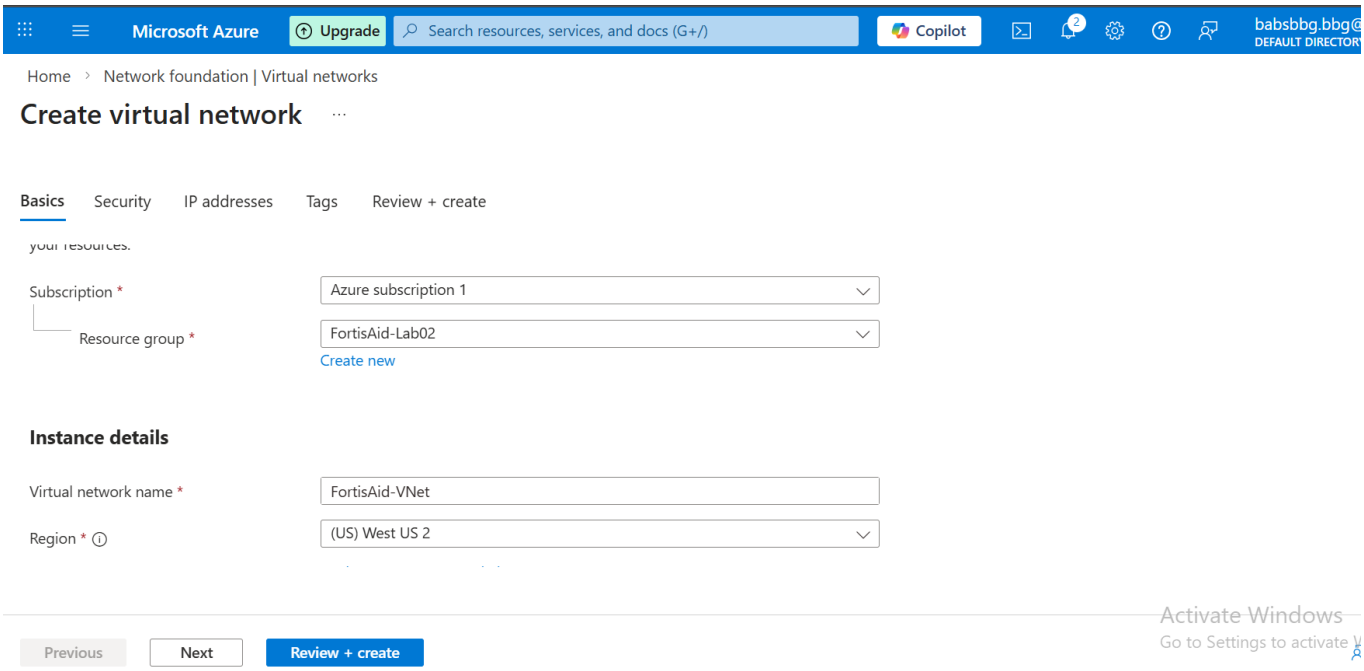
1. Azure Portal → Search **Resource groups**
2. Click **+ Create**
3. Configure:
   - Resource group: `FortisAid-Lab02`
   - Region: **West US 2**
4. Review + create → Create

Verification: RG appears in list.

Step 2 — Create Virtual Network + Subnet

1. Search **Virtual networks** → **+ Create**
2. Basics:
   - RG: `FortisAid-Lab02`
   - Name: `FortisAid-VNet`
   - Region: **West US 2**
3. IP Addresses:
   - Address space: `10.0.0.0/16`
   - Subnet: `default` with range `10.0.0.0/24`
4. Review + create → Create

Verification: `FortisAid-VNet` exists and shows the subnet.



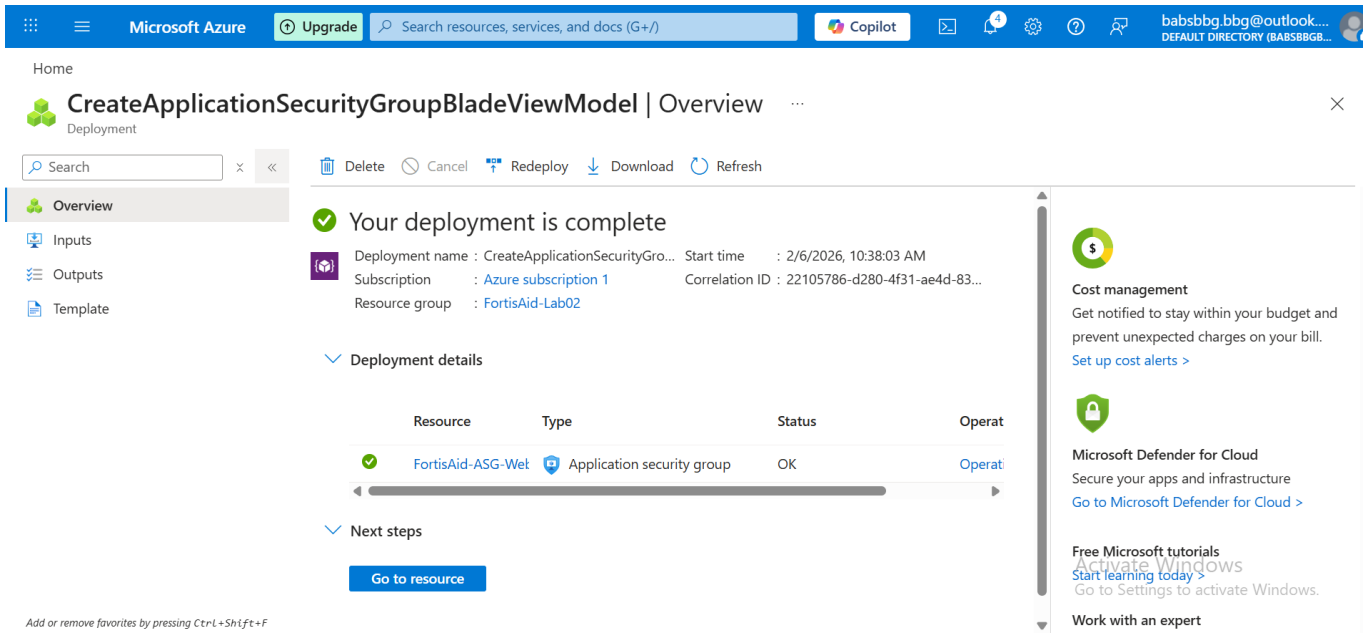
FortisAid-VNet overview showing address space and default subnet 10.0.0.0/24`

Step 3 — Create Application Security Groups (ASGs)

Create two ASGs:

ASG A (Web)

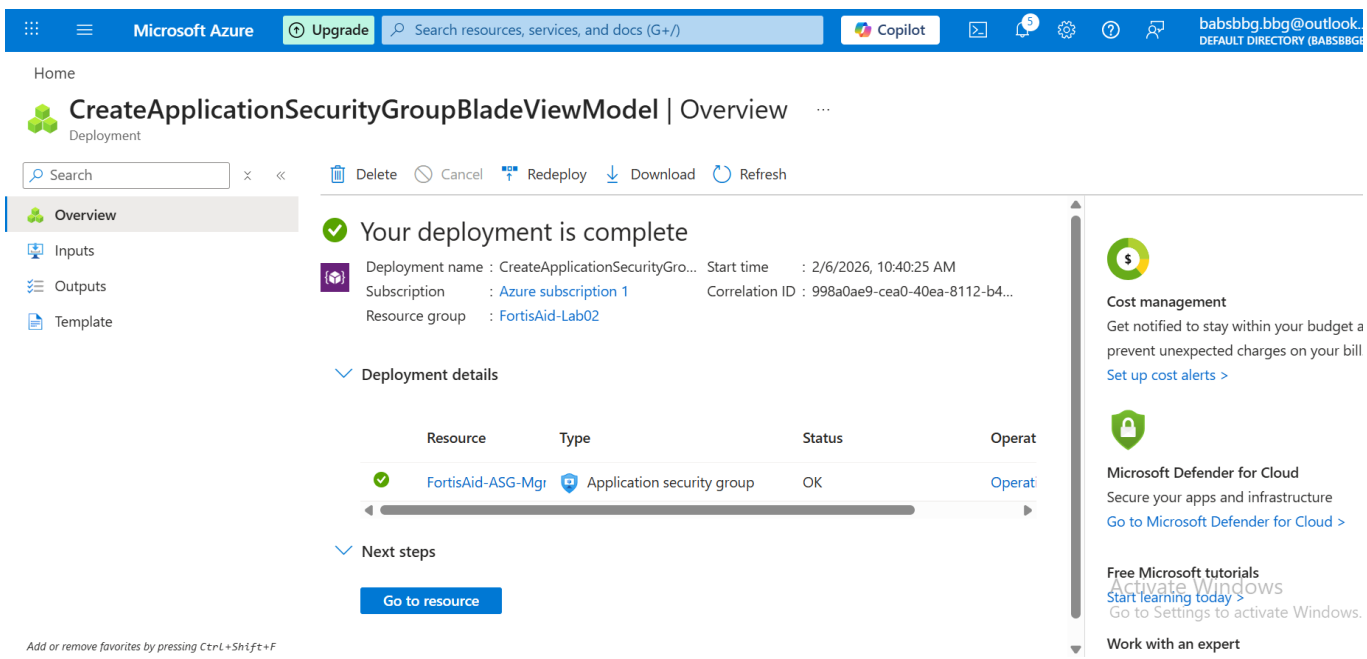
1. Search **Application security groups** → **+ Create**
2. RG: `FortisAid-Lab02`
3. Name: `FortisAid-ASG-Web`
4. Region: **West US 2**
5. Create



ASG B (Management)

Repeat:

- Name: `FortisAid-ASG-Mgmt`
  Verification: both ASGs appear in list.



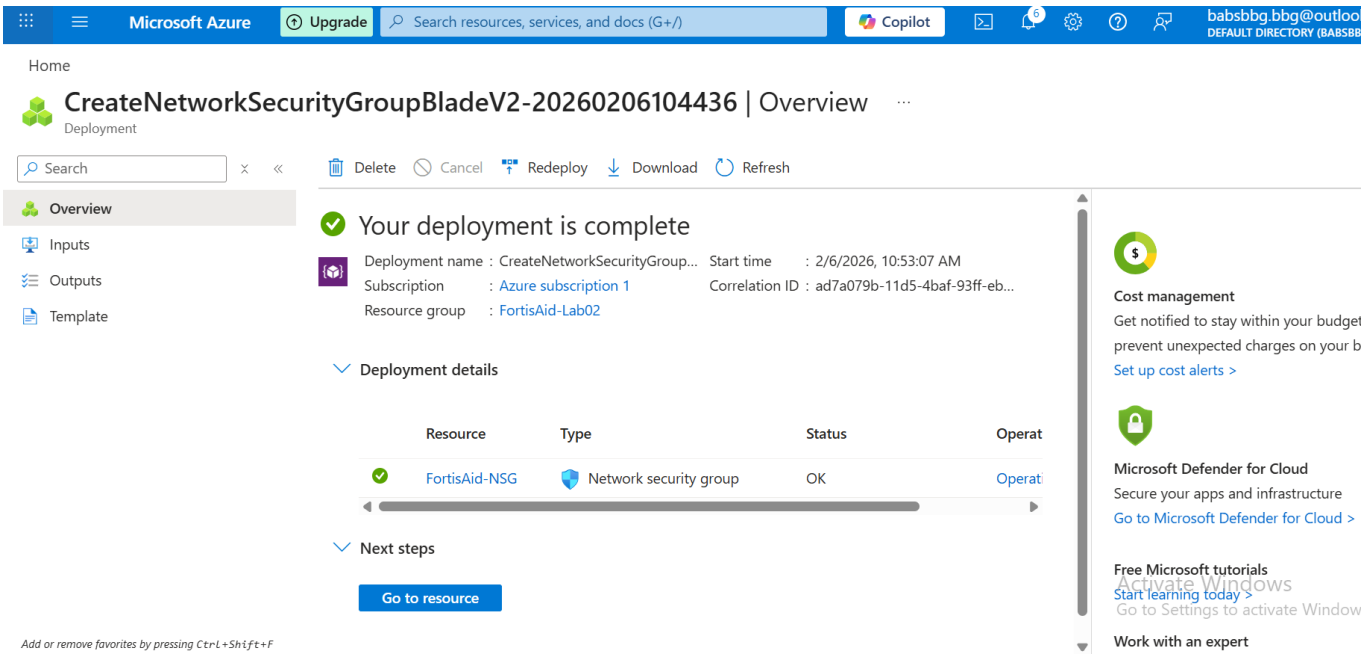
ASG list showing FortisAid-ASG-Web and FortisAid-ASG-Mgmt in West US 2`

Step 4 — Create Network Security Group (NSG)

1. Search **Network security groups** → **+ Create**
2. RG: `FortisAid-Lab02`
3. Name: `FortisAid-NSG`

4. Region: **West US 2**
5. Create

Verification: NSG appears.

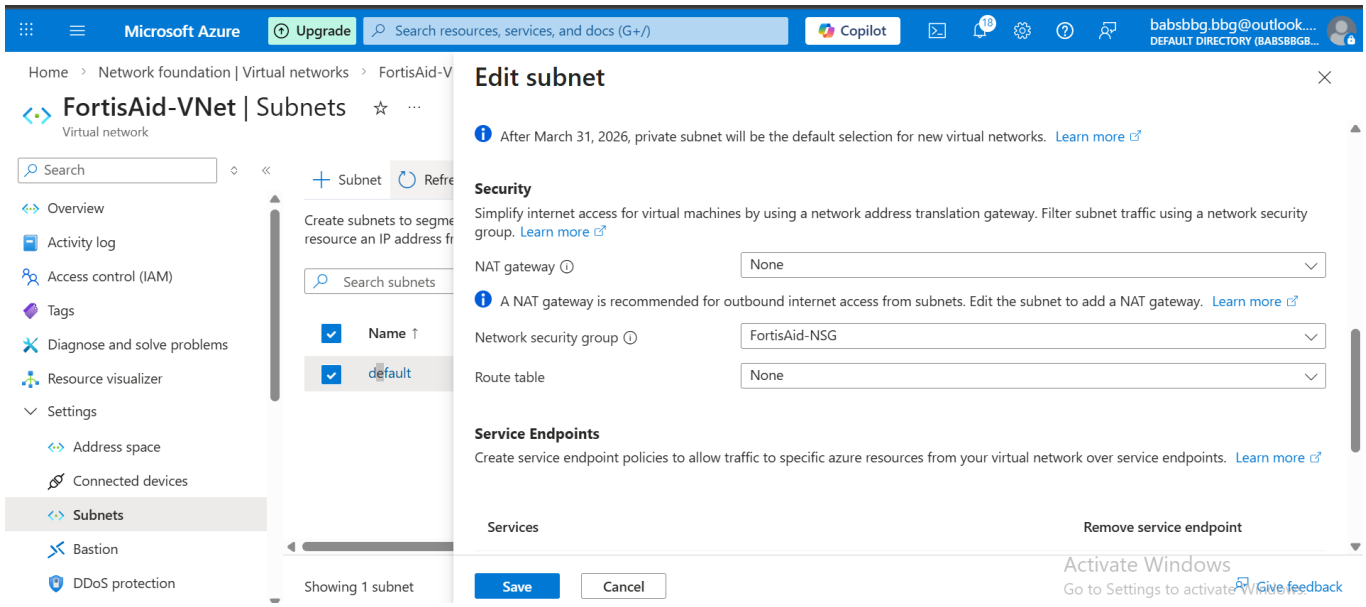


FortisAid-NSG overview showing Region = West US 2`

Step 5 — Associate NSG to the Subnet

1. Open **Virtual networks** → `FortisAid-VNet`
2. Left menu → **Subnets**
3. Click subnet: **default**
4. Under **Network security group**, click **Associate**
5. Select: `FortisAid-NSG`
6. Click **Save**

Verification: default subnet shows NSG attached.



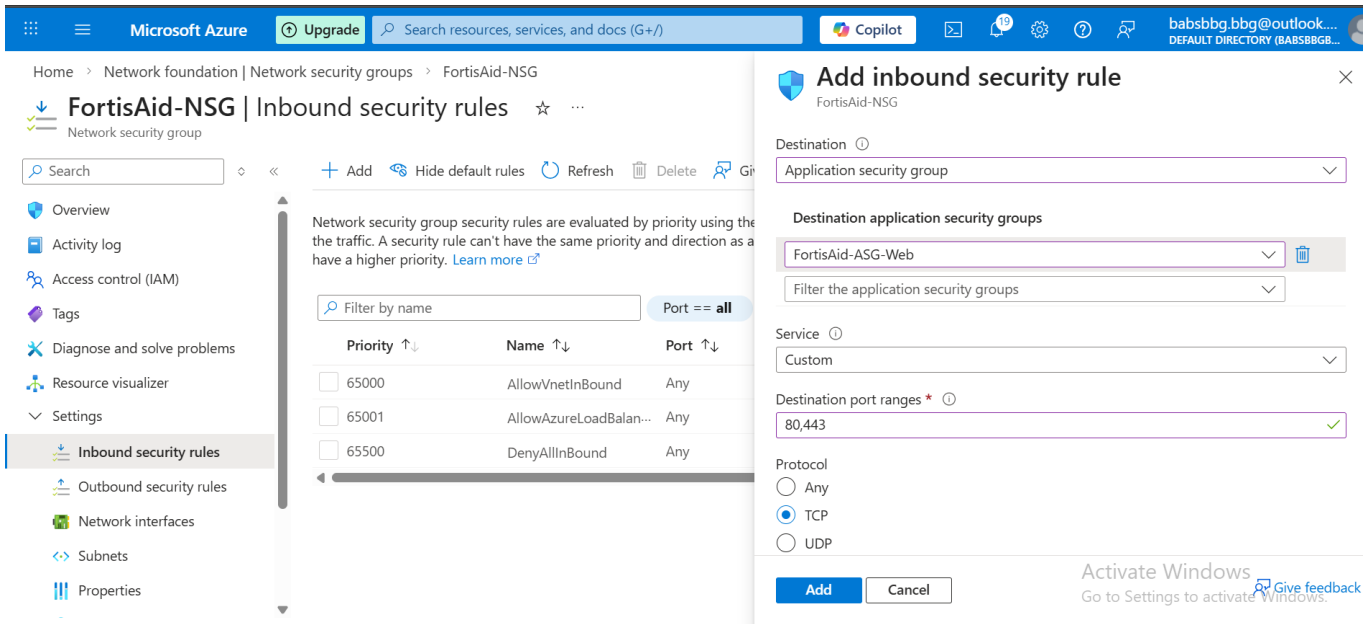
FortisAid-VNet > Subnets > default showing Network security group = FortisAid-NSG`

Step 6 — Create Inbound NSG Rules (ASG-targeted)

Open `FortisAid-NSG` → **Inbound security rules** → **+ Add**

Rule 1: Allow Web (HTTP/HTTPS) to Web ASG

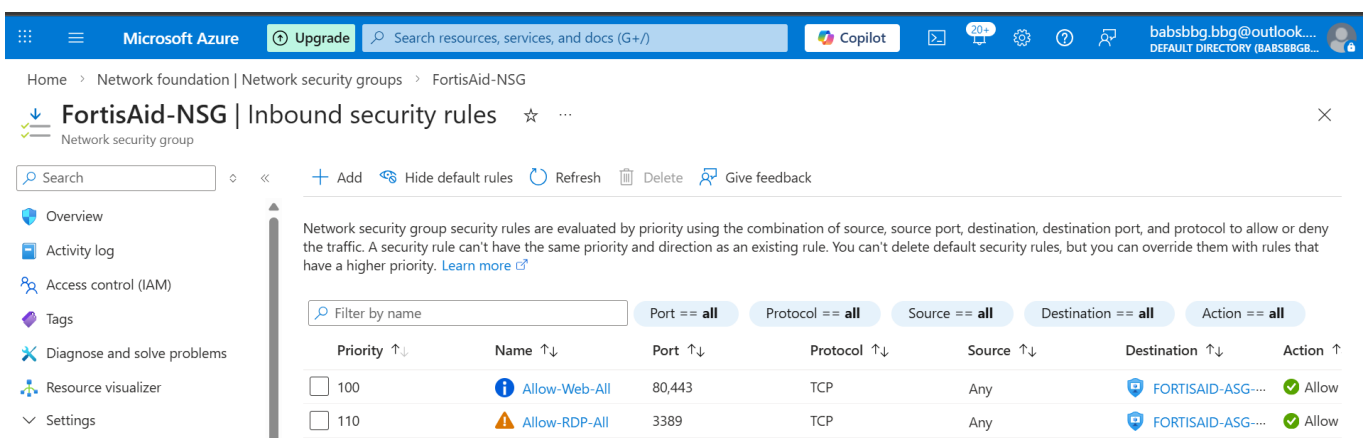
- Source: Any
- Destination: **Application security group** → `FortisAid-ASG-Web`
- Service: Custom
- Destination ports: `80,443`
- Protocol: TCP
- Action: Allow
- Priority: `100`
- Name: `Allow-Web-All`



Rule 2: Allow RDP to Mgmt ASG

- Source: Any
- Destination: **Application security group** → `FortisAid-ASG-Mgmt`
- Service: Custom
- Destination ports: `3389`
- Protocol: TCP
- Action: Allow
- Priority: `110`
- Name: `Allow-RDP-All`

Verification: both rules appear above defaults.



FortisAid-NSG inbound rules showing Allow-Web-All (80,443) and Allow-RDP-All (3389)`

Exercise 2: Deploy VMs and Test Network Filters

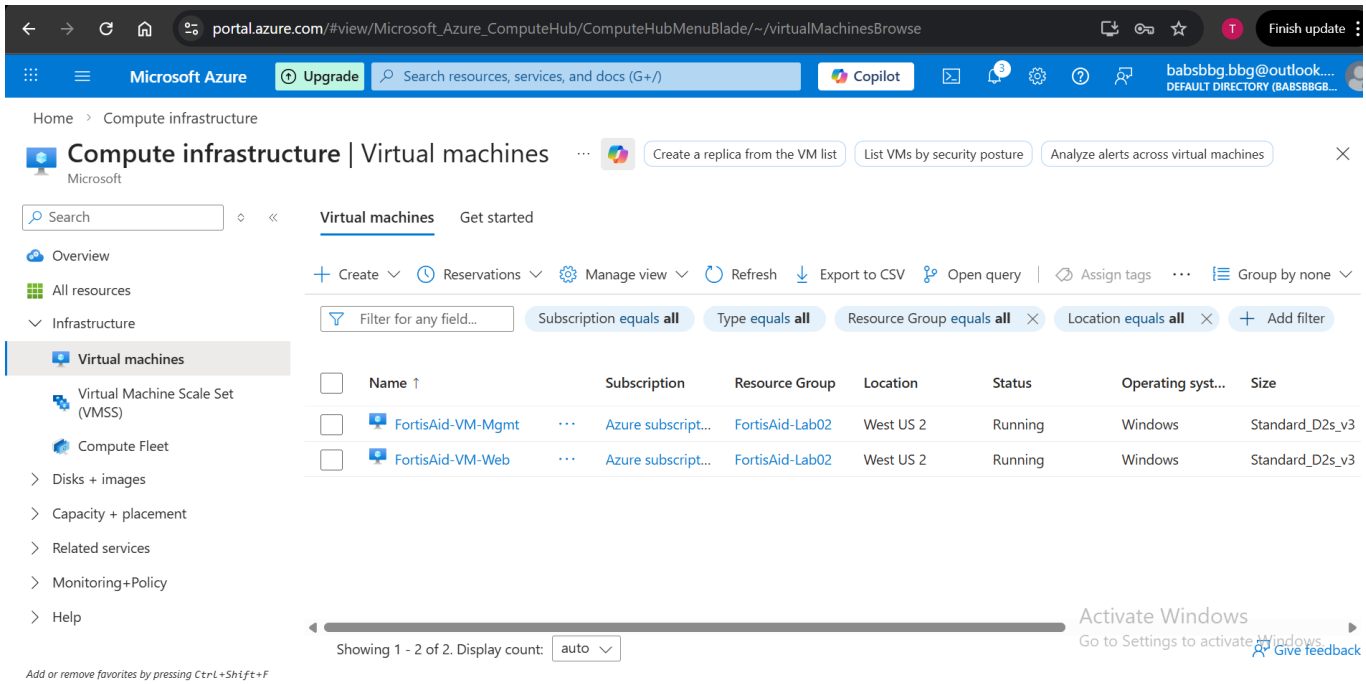
Step 7 — Create Web VM (no public inbound ports)

1. Search **Virtual machines** → **+ Create**
2. Basics:
   - RG: `FortisAid-Lab02`
   - Name: `FortisAid-VM-Web`
   - Region: **West US 2**
   - Availability: **No infrastructure redundancy required**
   - Image: Windows Server 2022 Datacenter (Gen2)
   - Size: B2s / D2s v3 (any available)
   - Username: `Student`
   - Password: your secure one
   - Public inbound ports: **None**
3. Networking:
   - VNet: `FortisAid-VNet`
   - Subnet: `default`
   - NIC NSG: **None**
4. Create

Step 8 — Create Mgmt VM (same network)

Repeat creation, but:

- Name: `FortisAid-VM-Mgmt`

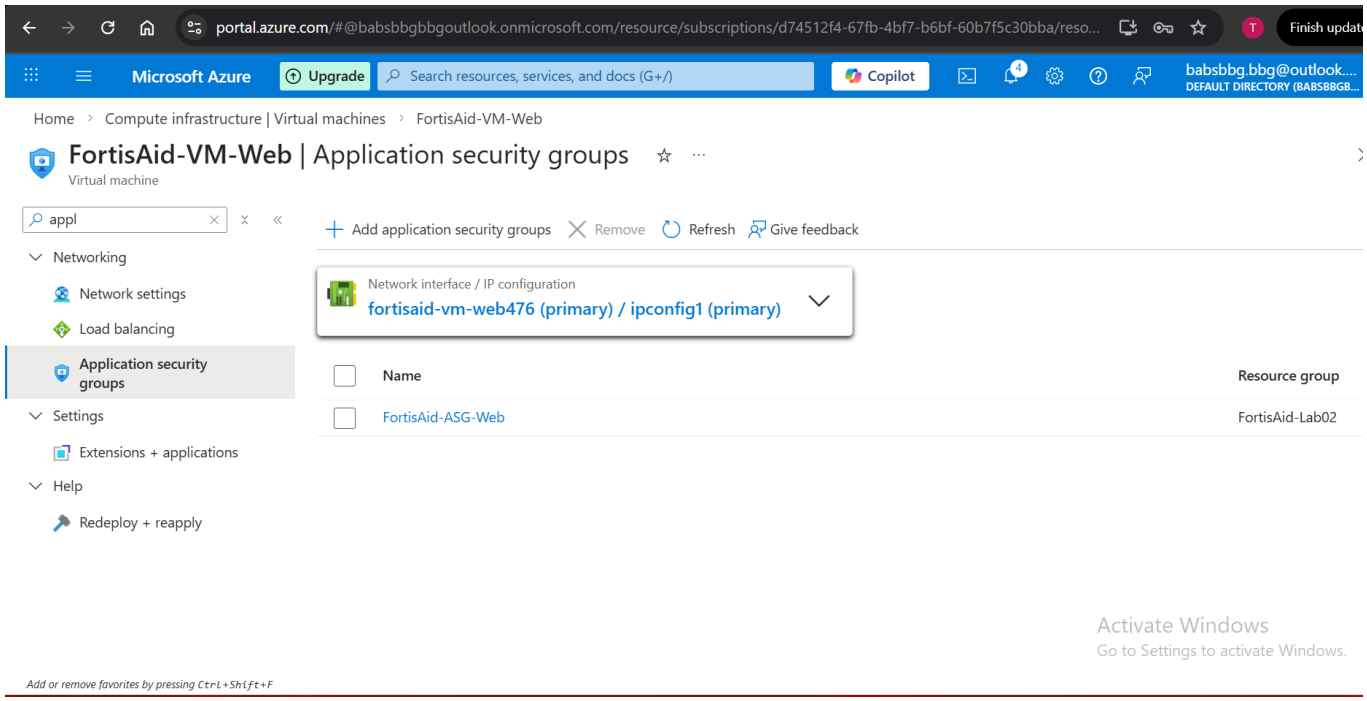


FortisAid-VM-Web & Mgmt overview showing Running and VNet/Subnet

Step 9 — Assign ASGs to NICs

Web VM → Web ASG

1. Open `FortisAid-VM-Web`
2. Networking → **Network settings**
3. **Application security groups** → Add
4. Select: `FortisAid-ASG-Web`
5. Save

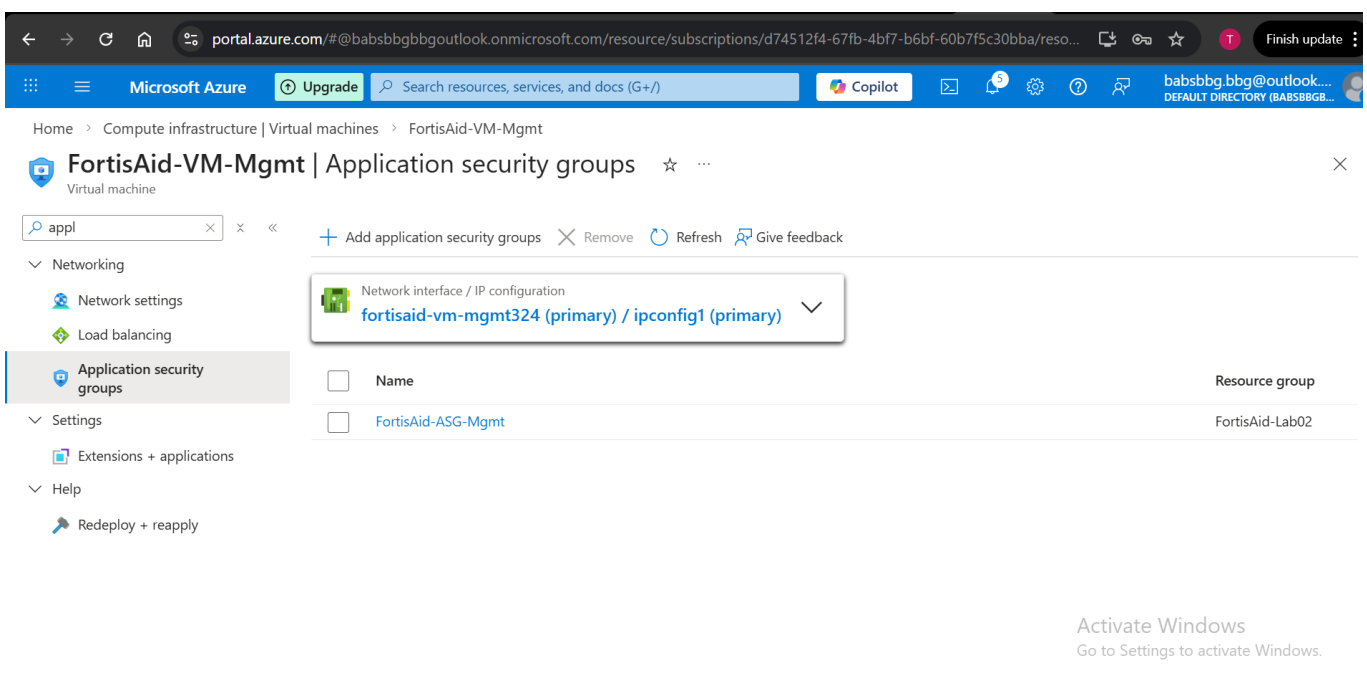


FortisAid-VM-Web networking showing ASG = FortisAid-ASG-Web`

Mgmt VM → Mgmt ASG

Repeat for `FortisAid-VM-Mgmt`:

- Select: `FortisAid-ASG-Mgmt`



FortisAid-VM-Mgmt networking showing ASG = FortisAid-ASG-Mgmt

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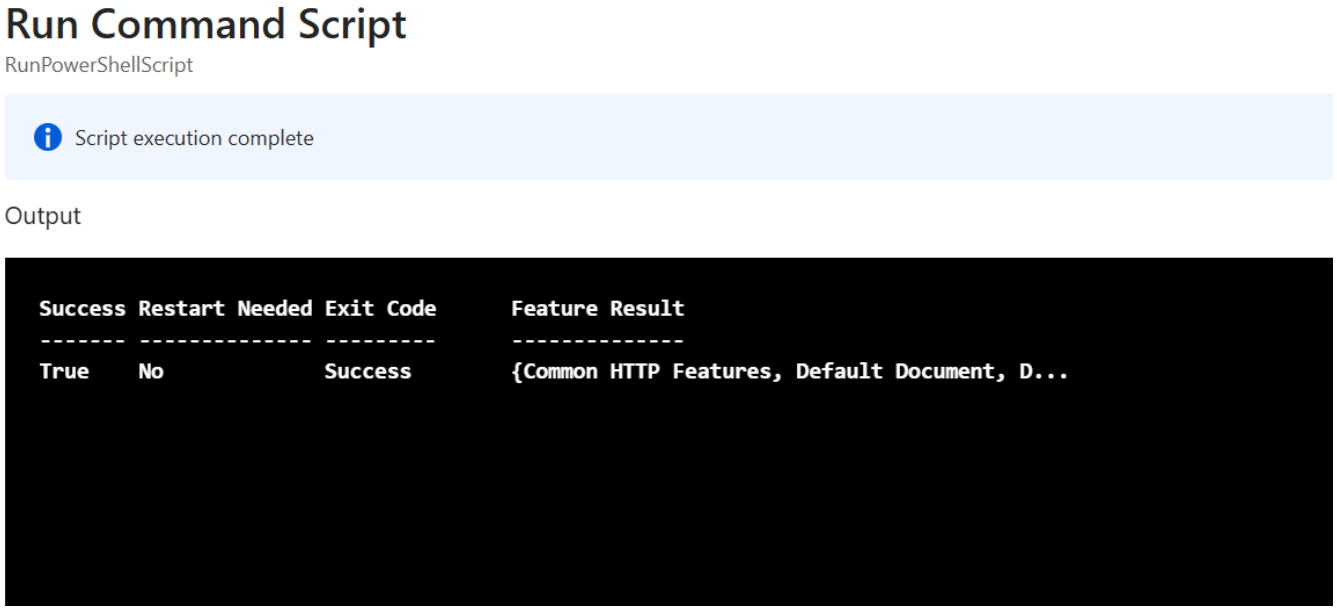
Step 10 — Install IIS on Web VM

1. Open `FortisAid-VM-Web`
2. **Run command** → RunPowerShellScript
3. Run:

```
Install-WindowsFeature -Name Web-Server -IncludeManagementTools
```

Verification: command returns success.

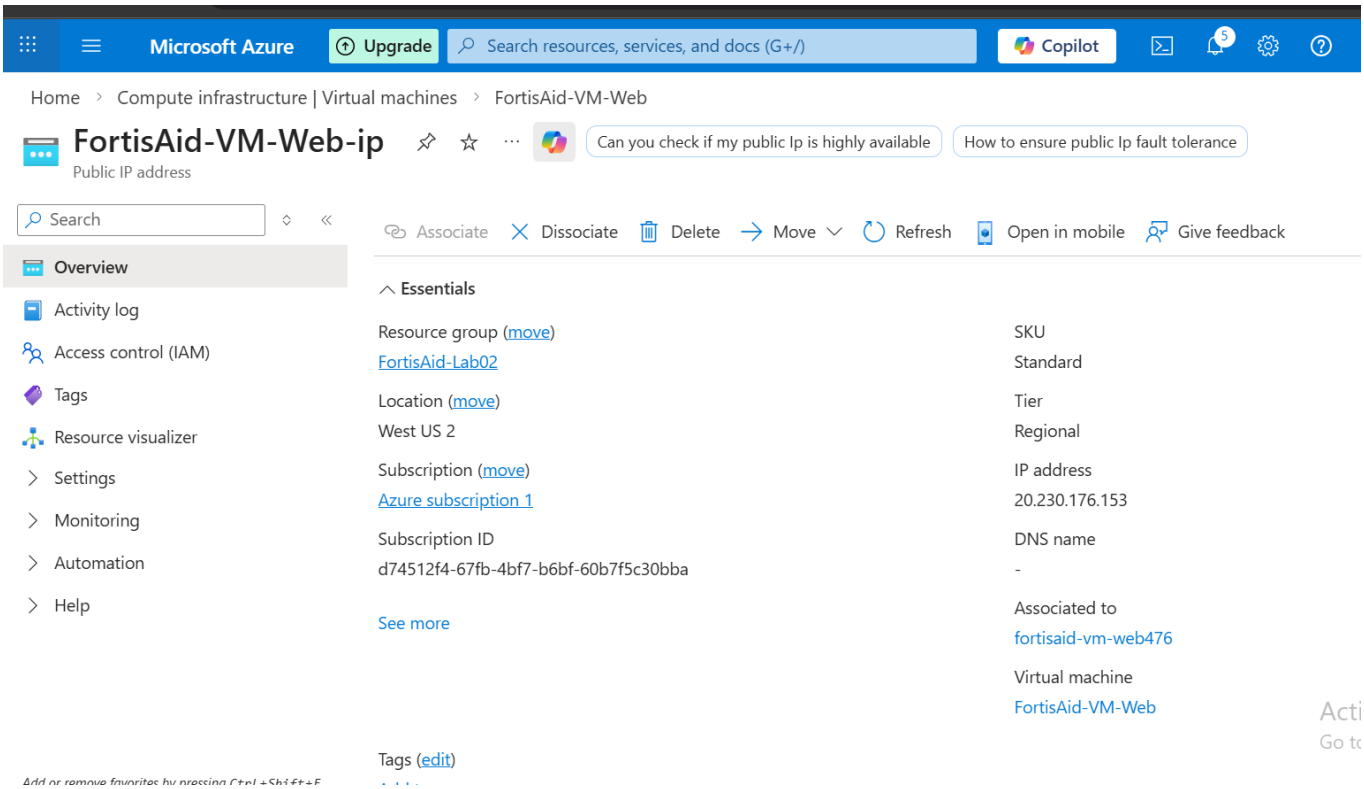


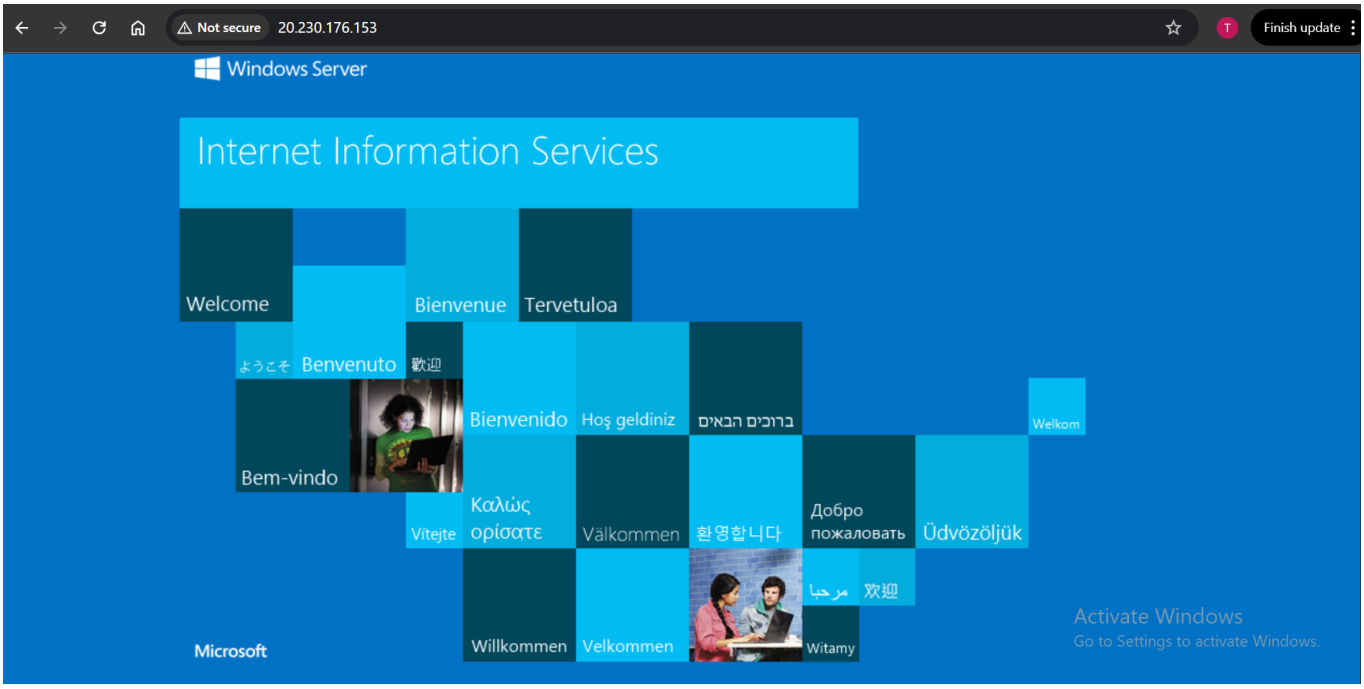
Run command output showing IIS installation succeeded

Step 11 — Testing the Network Security Controls

Test A: HTTP to Web VM (should WORK)

1. Ensure Web VM has a Public IP (VM Overview → Public IP)
2. Open browser:
   - `http://<WebVM-PublicIP>`
3. Expected: IIS default page loads





Browser showing IIS default page using FortisAid-VM-Web public IP

Test B: RDP to Mgmt VM (should WORK)

1. `FortisAid-VM-Mgmt` → Connect → RDP
2. Download RDP file and connect
3. Expected: RDP login succeeds

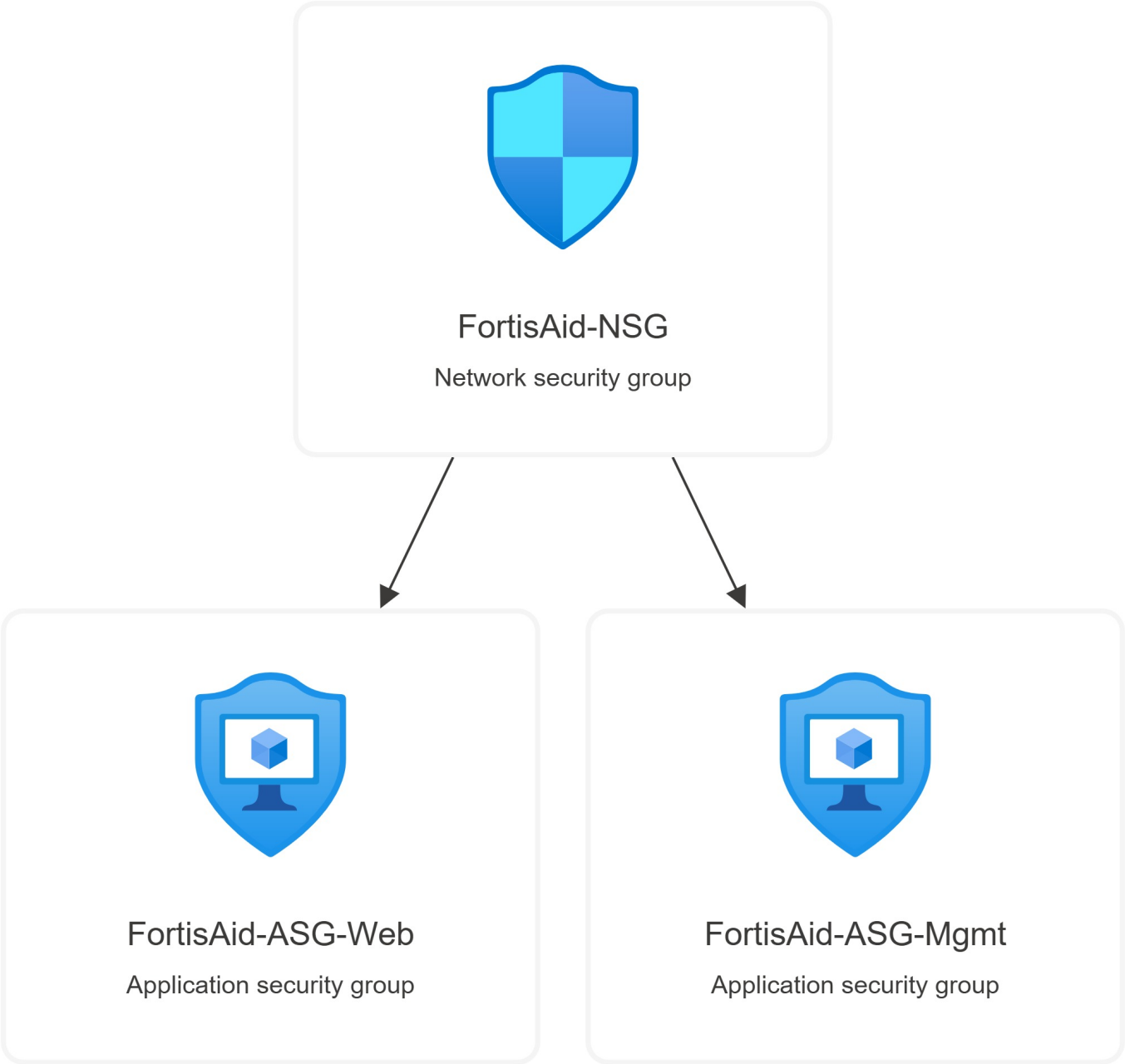
Test C: RDP to Web VM (should FAIL)

1. `FortisAid-VM-Web` → Connect → RDP
2. Attempt connection
3. Expected: connection fails / times out (blocked)

Final Step: Network and Application Security Groups Diagram

Create a diagram that shows:

- `FortisAid-VNet (10.0.0.0/16)`
- Subnet `default (10.0.0.0/24)`
- Subnet-level NSG `FortisAid-NSG`
- ASGs: `FortisAid-ASG-Web`, `FortisAid-ASG-Mgmt`
- VMs: `FortisAid-VM-Web` (IIS), `FortisAid-VM-Mgmt`
- Allowed flows: TCP 80/443 → Web ASG; TCP 3389 → Mgmt ASG
- Blocked flow: TCP 3389 → Web VM



NSG + ASG architecture diagram for FortisAid Lab 02